

← Main menu

Deploy Agentspace

Course - 2 hours ✓ Complete

🏠 Course overview

Deploy Agentspace ^

✓ Deploy and query Google Agentspace

✓ Deploy Google Agentspace with Data Stores and an action: Challenge Lab

Your Next Steps ^

✓ Course Badge

Course Survey

Course > Deploy Agentspace >

Quick tip: Review the prerequisites before you run the lab >

Start Lab

01:30:00

Deploy Google Agentspace with Data Stores and an action: Challenge Lab

Lab 1 hour 30 minutes No cost Intermediate

★★★★★

This lab may incorporate AI tools to support your learning.

Lab instructions and tasks

—/100

Challenge Lab Overview

Objective

Setup and requirements

Challenge Scenario

Your challenge

Task 1. Add content to Google Drive and Google Calendar

Task 2. Configure Agentspace Identity Provider and Create Google Drive and Google Calendar Data Stores

Task 3. Create an OAuth Consent Screen and Configure the Gmail Action

Task 4. Query your Agentspace Assistant

Challenge Lab Overview

< Previous

Thanks for reviewing this lab.

Next >

set of tasks - you figure out how to complete them on your own! An automated scoring system (shown on this page) provides feedback on whether you have completed your tasks correctly.

When you take a Challenge Lab, you will not be taught Google Cloud concepts. You will need to use your skills to assess how to build the solution to the challenge presented. This lab is only recommended for students who have those skills. Are you up for the challenge?

Objective

In this challenge lab, you will demonstrate your ability to deploy Google Agentspace and set up data stores and actions.

Note: To avoid confusion between your professional Google Identity and any other

When you take the lab, you will be logged in to the lab. After starting the lab, right-click on the "Open Google Cloud console" button and select "Open link in incognito window".

Setup and requirements

Before you click the Start Lab button

Read these instructions. Labs are timed and you cannot pause them. The timer, which starts when you click **Start Lab**, shows how long Google Cloud resources will be made available to you.

This Qwiklabs hands-on lab lets you do the lab activities yourself in a real cloud environment, not in a simulation or demo environment. It does so by giving you new, temporary credentials that you use to sign in and access Google Cloud for the duration of the lab.

What you need

- Access to a standard internet browser (Chrome browser recommended).
- Time to complete the lab.

Note: If you already have your own personal Google Cloud account or project, do not use it for this lab.

Note: If you are using a Pixelbook, open an Incognito window to run this lab.

How to start your lab and sign in to the Google Cloud Console

1. Click the **Start Lab** button. If you need to pay for the lab, a pop-up opens for you to select your payment method. On the left is a panel populated with the temporary credentials that you must use for this lab.

[Open Google Console](#)

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

Username
google2727832_student@qwiklabs.net

Password
k58CZxsxMZ

GCP Project ID

2. Copy the username, and then click **Open Google Console**. The lab spins up resources, and then opens another tab that shows the **Sign in** page.

Google

Sign in

Use your Google Account

Email or phone

[Forgot email?](#)

Tip: Open the tabs in separate windows, side-by-side.

If you see the **Choose an account** page, click **Use Another Account**.

Google

Your.email@gmail.com

google1381214_student@qwiklabs.net
Signed out

Use another account

3. In the **Sign in** page, paste the username that you copied from the Connection Details panel. Then copy and paste the password.

Important: You must use the credentials from the Connection Details panel. Do not use your Qwiklabs credentials. If you have your own Google Cloud account, do not use it for this lab (avoids incurring charges).

4. Click through the subsequent pages:

- Accept the terms and conditions.
- Do not add recovery options or two-factor authentication (because this is a temporary account).
- Do not sign up for free trials.

After a few moments, the Cloud Console opens in this tab.



Challenge Scenario



Here is a company overview as provided on the Cymbal Shops's website.

Remember, green energy, green earnings.

Founded in 1974, Cymbal Shops started out as Cymbal Air, selling AC systems manufactured by Cymbal Group in Minnesota and neighboring states. The company quickly expanded into domestic merchandise in order to satisfy the need for quality products at an affordable price in the midst of a recession.

Cymbal Shops' broad product assortment was once a benefit - capturing planned purchases and last minute splurges. In recent years, however, the company has struggled to adapt to the acceleration in e-commerce. Digitally native companies are on the rise and Cymbal Shops must implement significant changes in order to keep pace and maintain relevance. Today, Cymbal Shops operates 714 stores across North America and reported \$15 billion in revenue in 2019. They currently employ 80,400 employees across the United States and Canada.

Cymbal Shops is a digitally transforming legacy retailer.

It is inspired by clients like: Bed Bath & Beyond, Best Buy, Home Depot, Nordstrom.

Your challenge

You are an engineer working for Cymbal Shops, tasked with deploying Agentspace for the organization to help speed up knowledge exchange.

In this challenge lab, you will demonstrate your ability to deploy Google Agentspace and set up data stores and actions.

Your challenge includes the following tasks:

- Configure AI Applications authentication.
- Create Google Drive & Google Calendar data stores.
- Create an Agentspace app.
- Use the AI assistant to find information, generate insights, and take action by sending an email.

In this section, you will download files from a Cloud Storage bucket, upload them to a Google Drive folder, convert a document to Google Docs, and create a Google Calendar event. This prepares demo data for use in subsequent lab activities.

1. Navigate to **Cloud Storage** > **Buckets** and click on the existing bucket with the name: **GCP Bucket**.
2. Download the files listed below to your local system to use as demo data:
 - Cymbal Shops Q4 Sales Data by City.xlsx
 - Cymbal Shops Sales Analysis Report.docx
3. In the Incognito window where you have logged into the Google Cloud console, create a new tab and navigate to drive.google.com.
4. Create a new folder in your Qwiklab student account's drive and name it **Shop Name**.
5. Upload the documents you downloaded from Cloud Storage to the folder you just created in Google Drive named **Shop Name**.

The screenshot shows the Google Drive interface. At the top, the breadcrumb navigation reads 'My Drive > Cymbal Drive'. Below this, there are filters for 'Summarize this folder', 'Type', 'People', 'Modified', and 'Source'. A table lists the contents of the 'Cymbal Drive' folder:

Name	Owner	Last modified	File size
Cymbal Steeps 04 Sales Data by City.xlsx	me	7/26 PM '20	19 KB
Cymbal Steeps Sales Analysis Report.docx	me	1/27 PM '20	10 KB

6. In a new tab in your Incognito window, navigate to calendar.google.com. Create a calendar event named **Event Name** with a start time at least one hour from now.

Output:

Sales Data Review Meeting

☐ All day Does not repeat

Event details Find a time **Guests**

Add Google Meet video conferencing

Add location
 Notification ▾ 10 minutes ▾ X

Add notification

student c/c%4d50
 ☒ Yes
 ☐ No

Guest permissions
☒ Mutually event
☒ Invite others
☒ See guest list

View ▾ Default visibility ▾

Add description

Click *Check my progress* to verify the objective.

Add content to Google Drive and Google Calendar

Check my progress

Task 2. Configure Agentspace Identity Provider and Create Google Drive and Google Calendar Data Stores

created data stores.

1. Navigate to AI Applications and **enable the API**.
2. Configure AI Applications to use **Google Identity** as your identity provider for the `global` location.
3. Create a **Google Drive** data store named `Google Drive Data Store Name` with access to `All` folders and drives.
4. Create a **Google Calendar** data store named `Google Calendar Data Store Name`.
5. Create an **Agentspace** app named `Agentspace App` of type **Search + Assistant** and company named `Company Name` with access to the data stores created above.

Output:

Apps > agentspace-app-cymbal > Data stores

Selected data stores

Filter Enter property name or value	
Name	Type
cymbal-google-drive	Drive
cymbal-google-calendar	Calendar

Click **Check my progress** to verify the objective.

Configure Agentspace to search Google Drive and Google Calendar Data Stores

Check my progress

Task 3. Create an OAuth Consent Screen and Configure the Gmail Action

enabling user authorization. You will also create an OAuth client and add a Gmail action to your Agentspace app, allowing the AI assistant to generate email drafts. In order for the **AI assistant** to take actions on a user's behalf, the user must grant access to your Agentspace app.

1. In the **Google Auth Platform**, create an OAuth consent screen for your app. Name it `Google Auth App Name`. It should be intended for an **Internal** Audience.
2. Create an OAuth Client of type **web application** named `Agentspace Client` with the required Agentspace `Authorized redirect URI`.
3. Add a **Google Gmail action** within your Agentspace app to enable the AI assistant to generate email drafts from Agentspace. Provide the Action connector name `Gmail Action`.

Output:

Active Actions CREATE	
Filter	
Action Name	Source
send_email	Google Gmail

Click **Check my progress** to verify the objective.

Create an OAuth Consent Screen and Configure the Gmail Action

Check my progress

Task 4. Query your Agentspace Assistant

In this section, you will test the integration by querying the assistant about sales data and generating an email based on the analysis report.

and generating an email based on the analysis report.

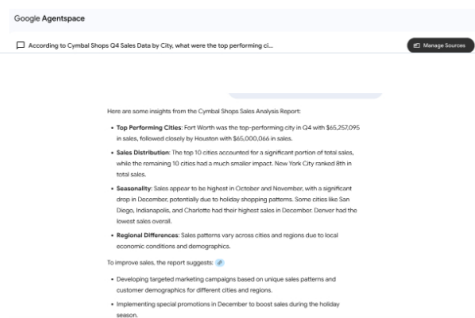
1. You may need to give the data store some time to ingest the data in the drive.
Take a 5 to 10 minute break and return to the next step.
2. From the app's Integration menu, find the **web app link** and click **Open**.
3. Using the primary search bar, ask: **According to Cymbal Shops Q4 Sales**

Output:



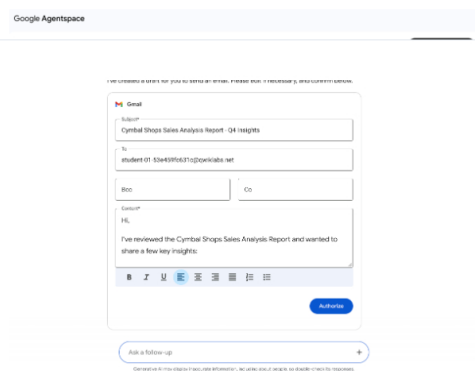
4. Using the primary search bar, ask: **What are some insights from the Sales Analysis Report?**

Output:



5. Using the primary search bar, ask: **Send an email to [INSERT_YOUR_EMAIL_ADDRESS_HERE] with high-level insights from the Cymbal Sales Analysis Report. Offer to discuss it further.** (If the output does not appear as a Gmail action you can Authorize, as pictured below, send the request to the assistant again. If it still doesn't work, the action may need a little more time to become active, so try again in a few minutes.)

Output:



6. When the action dialog appears, click **Authorize** to open the OAuth Consent Screen you configured, and authorize it with your Qwiklabs student account.
7. Once the application is authorized, click **Send** to send the email!

Congratulations!

In this lab, you've demonstrated your ability to deploy Google AgentSpace to ease knowledge exchange throughout an organization.

Manual Last Updated April 02, 2025

Lab Last Tested April 02, 2025

Copyright 2023 Google LLC All rights reserved. Google and the Google logo are trademarks of Google LLC. All other company and product names may be trademarks of the respective companies with which they are associated.

